AE 498 Planetary Defense, HW3

Problem (100 points)

This problem is based on an actual table-top exercise used in the 2025 Planetary Defense Conference, see Figure 1. Your **tasks**:

1. (10 points) Read through and summarize Epoch 1 of the 2025 PDC table-top exercise! `https://cneos.jpl.nasa.gov/pd/cs/pdc25/`
2. (30 points) Perform Initial Orbit Determination on the astrometry of the potential impactor 2024PDC25 provided in Epoch 1. The link to the observation file is here: `https://cneos.jpl.nasa.gov/pd/cs/pdc25/2024pdc25.xml` The results should be a set of orbital elements at a given epoch.
3. (40 points) Refine your initial orbit using all available astrometry through differential correction / OD and create a best fit orbit with covariance matrix.
4. (20 points) Propagate the best fit orbit and covariance matrix and confirm the likely date of impact of 2024 PDC25 as April 24, 2041. Show that the region of uncertainty intersects with the position of the Earth at the time of impact!

Hints:

- Feel free to use existing software, but document and motivate your choices!
- You do not have to recreate the risk corridor on Earth.
- Spiceypy and planetary Ephemerides can help, especially spk/planets and lsk! `https://naif.jpl.nasa.gov/pub/naif/generic_kernels`

Please document your choices in your homework report! Please also upload any code you generated to GitHub and post a link to the code you have used in your write-up!

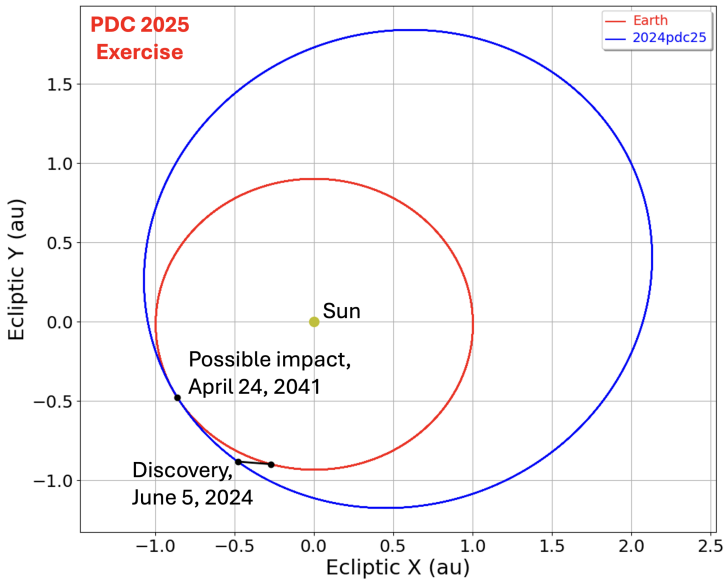

Figure 1: Good luck!